

[Language Translators]/ [Third]

Symbol table

[2025-2026]

In this lab we Create a new class for the symbol table that must be contain all the identifiers and attributes at first we must create a record then create an array of record to save the contents of the symbol tables and then print it on the output screen as follows: -

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Hours

[2 Hours]

IDNAME	IDTYPE	VALUE
A	Int	0
B	Int	0
X	Byte	0
Y	Byte	0

- The first class is ST_Rec that has
 - three var (IDname ,IDtype, and value) as a Strings and int respectively.
 - Method (create_ST_rec)
 - Method (getIDN) that return the IDname from record
 - Method (getIDT) that return the IDtype from record
- The second class is (ST_Table) that has
 - Define an array of the previous class of record as public (ST)
 - Define the index of array as public int (STsize)
 - Initialize the STsize to 0
 - Method (insert_ST) which is open new record in the ST_Rec class then call the (create_ST_rec) and inc the index by one.
 - Method (getID) that return the IDname from Table
 - Method (getT) that return the IDtype from Table
 - Method (member) which search for an ID if it available in the symbol table or not so it must be return true or false .
 - Method Run_ST that has an object of Look_up_Table, in this method we search for each ID in the LUT table and save it in the symbole table with its type and value.

J	I	Lexem	Type	IDNAME	IDTYPE	VALUE
-	0	DEF	RW			
i+1	1	Int	RW	a	Int	0
i+2	2	a	ID	b	Int	0
i+3	3	,	SP	x	Byte	0
i+4	4	b	ID	y	Byte	0
i+5	5	;	SP			

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i+i	6	Byte	RW
i+1	7	x	ID
i+2	8	,	SP
i+3	9	y	ID
i+4	10	:	SP
Stop j			

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Stop j

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Run_ST method: -

Loop i=(0→“Body”)

If lexeme == “int” or “byte”

Loop j=(i+1→”;”)

If (type==”ID” and not member(lexeme)

Then insert_ST(lexeme(j), lexeme(i),”0”)

Now we must return to i loop and complete checking

End j

End i

3- Now we must define an object from the ST_Table in the compiler class then call the Run_ST method and print the ST_table.

```
package compiler;
public class ST_Rec {
    private String IDName;
    private String IDType;
    private int Value;

    public void creat_ST_rec(String id,String t,int v){
        IDName=id;
        IDType=t;
        Value=v;}

    public String getIDN(){return(IDName);}
    public String getIDT(){return(IDType);}
}
```

```
package compiler;
public class ST_Table {

    public ST_Rec[] ST=new ST_Rec[100];
    public int STsize;
    ST_Table(){STsize=0;}

    public void insert_ST(String id,String t)
    {
        ST[STsize]=new ST_Rec();
        ST[STsize].creat_ST_rec(id, t,0);
        STsize++;
    }

    public String getID(int i)
    {
        return(ST[i].getIDN());}
}
```

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public String getT(int i)
return (ST[i].getIDT());}

public boolean member(String id){
for (int i=0;i<STsize;i++){
if (id.equals(ST[i].getIDN()))

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Hours [2 Hours]

```
public void runST(Look_up_table L) {
    int i,j;
    for (i=0;!("BODY".equals(L.getLex(i)));i++)
    {
        if ((L.getLex(i).equals("int"))||(L.getLex(i).equals("byte")))
        {
            for (j=i+1;!(L.getLex(j).equals(";"));j++)
            if ((L.getTyp(j).equals("ID"))&&!(member(L.getLex(j))))
                insert_ST(L.getLex(j),L.getLex(i));
            i=j;
        }
    }
    }/// end of runst
}
```

```
///// the sixth lab Symbol Table /////
package compiler;

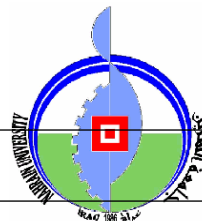
public class Compiler {
    public static void main(String[] args) {
        try{
            /////////////// Lexical ///////////////
            Look_up_table Lutt=new Look_up_table();
            Lexical L=new Lexical();
            L.run(Lutt);
            System.out.println("*****LUT*****");
            System.out.println("*****");
            for(int i=0;i<Lutt.size;i++)
                System.out.println(Lutt.getLex(i)+"    Type is    "+Lutt.getTyp(i));

            ///////////////Symbole Table//////////
            ST_Table St2=new ST_Table();
            St2.runST(Lutt);
            System.out.println("Symbole Table");
            System.out.println("*****");
            for (int m=0;m<St2.STsize;m++)
                System.out.println(St2.getID(m)+"    type is    "+St2.getT(m));

        }
        catch(Exception e){}
    }
}
```

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